**Course Code : ECT 357** **MQNR/RS – 23 / 1024**

## Sixth Semester B. Tech. ( Electronics and Communication Engineering ) Examination

## COMPUTER NETWORKS

Time : 3 Hours ] [ Max. Marks : 60

**Instructions to Candidates :—**

(1) All questions are compulsory.

(2) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (A) What are some of the factors that determine whether a communication system is a LAN or WAN ? 5(CO1)

   (B) What are headers and trailers and analyze how do they get added and removed ? 5(CO1)

2. (A) Compare and contrast a circuit-switched network and a packet-switched network. 5(CO3)

   (B) List three categories of Multiple Access Protocols. Discuss any one protocol in details. 5(CO3)

3. (A) In a stop-and-wait ARQ system, the bandwidth of the line is 1 Mbps, and 1 bit takes 20 ms to make a round trip. What is the bandwidth-delay product ? If the system data frames are 1000 bits in length, there is the underutilization of the link ? Justify. 5(CO4)

   (B) Discuss the relationship between STS and STM. What are the user data rates of STS-3, STS-9 and STS-12 ? Show how STS-9's can be multiplexed to create an STS-36. Is there any extra overhead involved in this type of multiplexing. 5(CO4)

4. (A) Explain Dijkstra Algorithm and also discuss the scenario where Dijkstra algorithm may not work. 5(CO3)

(B) Design a network for an organization which is granted with the block of IP address 16·0·0·0/8. The administrator wants to create 500 fixed-length subnets.

(a) Find the subnet mask.

(b) Find the number of addresses in each subnet.

(c) Find the first and last addresses in subnet 1.

(d) Find the first and last addresses in subnet 500. 5(CO5)

5. (A) Compare the TCP header and the UDP header. List the fields in the TCP header that are missing from UDP header. Give the reason for their absence. 5(CO4)

(B) What does HTTP stand for and what is its function ? How is HTTP related to WWW ? 5(CO2)

6. (A) Explain Symmetric-Key Cryptography. Using Caesar ciphering with key = 15, encrypt the message "HELLO". 5(CO2)

(B) For a RSA algorithm with p = 3 and q = 11, evaluate public key and private key. Also obtain the ciphertext for the plaintext 'HELLO'. 5(CO2)